

Fleet Route Optimization Report

Generated (UTC): 2026-08-28T17:12:23.681837+00:00 | Use Case: GENERIC
Algorithm: QPSO-VRP | Data Source: FALLBACK TRAFFIC

1. Executive Summary

Total Fleet Distance:	777.24 km
Total Travel Time:	15.54 hrs
Estimated Fuel Cost:	Rs.6217.92 (assumed Rs.8.0/km)
Traffic Delay Estimate:	N/A (fallback mode)
On-Time Delivery Rate:	0.0%
Vehicles / Customer Stops:	1 vehicles / 1 stops

2. Vehicle Route Manifests

Vehicle #1 - Distance: 777.24 km | Duration: 15.54 hrs

Seq	Stop Name	Arrival (h)	Leg Dist (km)	Traffic	Status
1	Hyderabad	8.0h	0.0	UNKNOWN	-
2	Mumbai	23.54h	777.24	UNKNOWN	LATE

3. Traffic & Congestion Analysis

Segment Breakdown: High Congestion: 0 | Medium: 0 | Low: 0
Note: Fallback free-flow traffic model used. Figures assume ideal road conditions.

4. Optimization Performance

Total Iterations: 100 | Initial Cost: 17566.85 | Final Optimal Cost: 17566.85
Convergence Rate: None | Quantum Tunneling Events: 0 | Runtime: 1.1856s

5. Recommendations & Operational Notes

- Total distance, time, and cost figures above summarize this specific optimization run and are based on the data source noted in metadata.
- Compare the optimization_performance section across repeated runs to gauge consistency of the solver on your data.
- Data source for this report: fallback traffic data, as noted in metadata.data_source. Fallback-mode figures assume free-flow conditions and will understate real-world delay.